

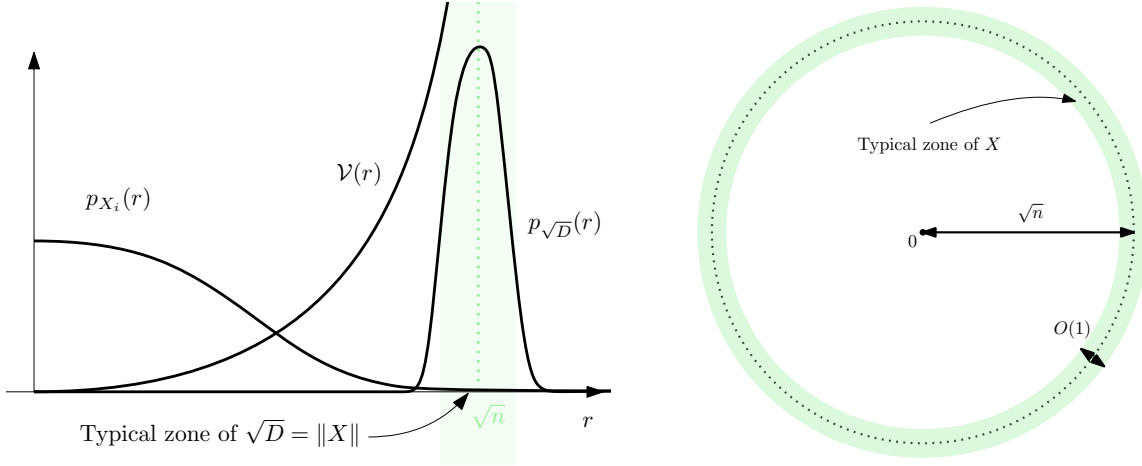


Figure 5.3: **(Left)** Schematic view of the density of one marginal X_i , of the norm $\sqrt{D}$ and of the volume $V(r)$ depending on the distance to 0: r . **(Right)** Schematic view of the distribution of $X \sim \mathcal{N}(0, I_n)$, it concentrates around the sphere $\sqrt{n}\mathbb{S}^{n-1}$.

measure on $\mathbb{R}$, one can therefore, thanks to Radon-Nikodym Theorem, introduce its density that satisfies for any interval $[a, b]$ of $\mathbb{R}$ with $0 \leq a \leq b$:

$$\begin{aligned} \int_a^b p_D(r) dr &= \mathbb{P}_D([a, b]) = \mathbb{P}(D \in [a, b]) \\ &= \int_{a \leq \|x\|^2 \leq b} p_X(x) d\lambda_n(x) \\ &= \frac{1}{(2\pi)^{n/2}} \int_{a \leq x_1^2 + \dots + x_n^2 \leq b} e^{-\frac{1}{2}(x_1^2 + \dots + x_n^2)} dx_1 \dots dx_n. \end{aligned}$$

With the use of n -dimensional spherical coordinates, some integral calculus (which is out of the scope of this course) can then allow us to show that one can choose:

$$p_D(r) = \frac{r^{n-1}}{(2\pi)^{n/2}} e^{-\frac{r^2}{2}}.$$

This density is the product of a volume term $\mathcal{V}(r) = \frac{r^{n-1}}{(2\pi)^{(n-1)/2}}$ and the marginal density $p_{X_i}(r) = \frac{1}{(2\pi)^{1/2}} e^{-\frac{r^2}{2}}$. Those two influences implies that the maximum is reached around n (see Figure 5.3 **(Left)** that provides an intuition). Indeed one can compute the mode (i.e. $\text{Argmax}_{r>0} p_D(r)$) followingly:

$$p'_D(r) = 0 \iff (n-1)r^{n-2}r^{n-1}e^{-\frac{r^2}{2}} = r^n e^{-\frac{r^2}{2}} \iff r = (n-1).$$

Note that the mode $n-1$ is not equal to the mean n but still it is very close. More details of this phenomenon will be given in the lecture about concentration of the measure.

3 From sampling to estimates

3.1 Sampling with inverse distribution function

Definition 5. Given a random variable $X : \Omega \rightarrow \mathbb{R}$, the cumulative distribution function – “the **cdf**” – of X , $F_X : \mathbb{R} \rightarrow [0, 1]$ is the mapping defined for all $x \in \mathbb{R}$ as:

$$F_X(x) = \mathbb{P}(X \leq x)$$

Lemma 5.9. For continuous random variables $X : \Omega \rightarrow \mathbb{R}$, there is a one-to-one correspondence between the density p_X (also called the probability density function – “the **pdf**”), and the cdf F_X :

$$\forall x \in \mathbb{R} : \quad F_X(x) = \int_{-\infty}^x p_X(t) dt \quad \text{and} \quad p_X(x) = F'_X(x).$$

Lemma 5.10. Given a random variable $U \sim \text{Unif}([0, 1])$ and a strictly increasing and on-to² (thus invertible) mapping $F : \mathbb{R} \rightarrow [0, 1]$ the cumulative distribution function of $F^{-1}(U)$ is exactly F :

$$F_{F^{-1}(U)} = F$$

Proof. Given $u \in \mathbb{R}$, let us simply compute:

$$\mathbb{P}(F^{-1}(U) \geq u) = \mathbb{P}(U \geq F(u)) = \int_0^{F(u)} dt = F(u).$$

□

This Lemma is of high importance because it allows to simulate all the classical random variables thanks to the “pseudo random number generator” which is a method present in all computers in order to sample independent random variables uniformly in $[0, 1]$. One then just has to apply F_X^{-1} to such samples to obtain independent samples of a given random variable $X : \Omega \rightarrow \mathbb{R}$.

In practice, more efficient methods like the “Box-Muller method” are generally employed to sample multidimensional random vectors. But this method is arguably too elaborate for a presentation in this course.

3.2 Rejection sampling

Let us consider a simple but representative example.

Example 5.11. One wants to sample a variable $Y : \Omega \rightarrow \mathbb{R}$ having a Gaussian truncated density:

$$f : y \mapsto \frac{e^{-y^2/2} \mathbb{1}_{[a, +\infty)}}{\int_a^{+\infty} e^{-t^2/2} dt},$$

for a given $a \in \mathbb{R}$. In the case of the Gaussian distribution, one can use precise numerical estimate of the erf function to compute $F_Y : t \mapsto \int_{-\infty}^t f(y) dy$. But let’s say that we do not have those numerical tools (or that the distribution that is truncated is more complex than the Gaussian) there still exists a naive way to sample Y which is the following:

1. Sample $X \sim \mathcal{N}(0, 1)$ (it has the density $t \mapsto \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$),
2. if $X \geq a$ accept it, otherwise reject it.

It can be shown that the set of the accepted samples following this procedure will indeed have as density f . However, if a is too big (let’s say $a = 4.5$) then most of the sample of X will satisfy $X \leq 4.5$, there is indeed approximately a chance out of 10000 that $X \geq 4.5$. Therefore this procedure looks highly inefficient, and we propose below a more relevant sampling method.

In the previous subsection, given a random variable $Y : \omega \mapsto \mathbb{R}$, one has to know F_X in order to sample X . As pictured in Example 5.11 it often happens in statistical problems that one just has partial information on the density of X and, more importantly, that there is no close-form for the integration of this density: F_X . Typically, one knows that the density f is proportional to a certain function $\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}_+$ ($\tilde{f} : y \mapsto e^{-y^2/2} \mathbb{1}_{[a, +\infty)}$ in Example 5.11). This proportional relation is denoted $f \propto \tilde{f}$, and that means that there exists a constant $C > 0$ such that $\forall x \in \mathbb{R} : f(x) = C \tilde{f}(x)$. Of course, in that case, since $\int p_X = 1$, one has $C = \frac{1}{\int \tilde{f}}$.

²Here it means that $F(\mathbb{R}) = [0, 1]$.

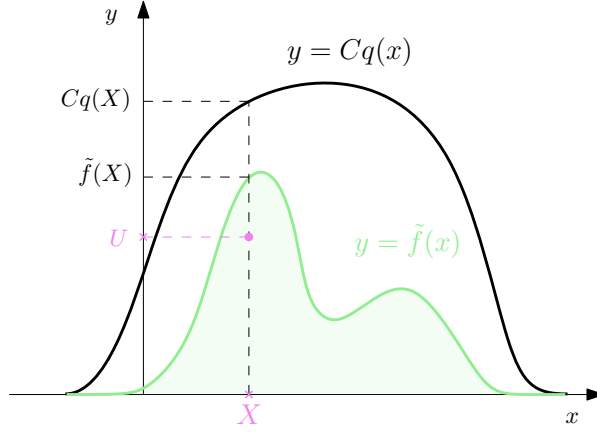


Figure 5.4: Two possible samples of U and X . $U \leq \tilde{f}(X)$ so we accept X .

Given a density $f \propto \tilde{f}$ (we have knowledge on $\tilde{f}$ but not on f) rejection sampling relies on the existence of an other density $q : \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a constant $C > 0$ satisfying:

$$\forall x \in \mathbb{R} : \quad f(x) \leq Cq(x).$$

Sample data set of the rejection sampling method is obtained iteratively following on each iteration those three steps:

1. sample x from the distribution X with density q ,
2. sample u from $U \sim \text{Unif}([0, Cq(x)])$,
3. if $u \leq \tilde{f}(x)$ keep the value x in the final sample data set (“Accept”) and if $u > \tilde{f}(x)$ reject the value x (see Figure 3.2).

Doing this iteration 5 times would be formalized with 5 independent random variables X_1, X_2, X_3, X_4, X_5 all having density q , 5 independent variables U_1, U_2, U_3, U_4, U_5 respectively dependent on X_1, X_2, X_3, X_4, X_5 since $\forall i \in [5], U_i \sim \text{Unif}([0, Cq(X_i)])$. Then if, for a particular sample $w_1, w_2, w_3, w_4, w_5 \in \Omega$, one has:

$$U(w_1) \leq \tilde{f}(X(w_1)), \quad U(w_2) > \tilde{f}(X(w_2)), \quad U(w_3) \leq \tilde{f}(X(w_3)), \quad U(w_4) \leq \tilde{f}(X(w_4)), \quad U(w_5) > \tilde{f}(X(w_5)),$$

the kept values would be $\{X(w_1), X(w_3), X(w_4)\}$. This selection is done under the condition $U \leq \tilde{f}(X)$ (i.e. under the event $\{w \in \Omega, U(w) \leq \tilde{f}(X(w))\}$), therefore, the density of the associated random variable is the conditional density:

$$x \mapsto p_X(x \mid U \leq \tilde{f}(X)).$$

Proposition 5.12. *The random variable output from rejection sampling has density f .*

The following proof relies on the notation for conditional densities. Since the computation rules are quite intuitive, we allow ourselves to present them here although the rigorous definitions of such objects will be provided in Section 4

Proof. One wants to compute:

$$p(Y = y \mid U \leq \tilde{f}(X)) = \frac{p(X = y, U \leq \tilde{f}(X))}{\mathbb{P}(U \leq \tilde{f}(X))} = \frac{p(U \leq \tilde{f}(y) \mid X = y)p_X(y)}{\mathbb{P}(U \leq \tilde{f}(X))}$$

By definition, $p_X(y) = q(y)$,

$$p(U \leq \tilde{f}(x_0) \mid X = y) = \frac{1}{Cq(y)} \int_0^{\tilde{f}(y)} dt = \frac{\tilde{f}(y)}{Cq(y)},$$

and:

$$\mathbb{P}(U \leq \tilde{f}(X)) = \int_{\mathbb{R}} p(U \leq \tilde{f}(x_0) | X = y) p_X(x) dx = \frac{\int \tilde{f}}{C}.$$

Finally one gets as expected:

$$p(Y = y | U \leq \tilde{f}(X)) = \frac{\tilde{f}(y)}{\int \tilde{f}} = f(y).$$

□

3.3 Monte Carlo estimation - problem setting

A common problem in statistical learning is to estimate expectations $\mu \equiv \mathbb{E}[h(X)]$ for a random variable $X : \Omega \rightarrow \mathbb{R}$ having a density f and a certain measurable mapping $h : \Omega \rightarrow \mathbb{R}$. The law of large numbers gives us:

Theorem 5.13 (Law of large numbers). *Given any sequence of i.i.d. random variables $X_1, X_2, \dots : \Omega \rightarrow \mathbb{R}$, for all $\varepsilon > 0$:*

$$\mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_1] \right| \geq \varepsilon \right) \xrightarrow{n \rightarrow \infty} 0.$$

Therefore a first naive idea is to use as estimator for $\mu = \mathbb{E}[X]$ the random variable $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$. If there were no computation limitation, Theorem 5.13 would ensure that for n big enough, $\hat{\mu}$ can be as close to μ as needed. However the big limitation of Theorem 5.13 is that it does not provides the speed of convergence, and we will see in Example 5.15 that the variance (and thus the speed) varies a lot depending on the estimator. Actually, X_1 is already an estimator for μ since $\mathbb{E}[X_1] = \mu$, however the big improvement with $\hat{\mu}$ is that (when the variance is bounded):

$$\mathbb{V}[\hat{\mu}] = \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}[X_i] = \frac{1}{n} \mathbb{V}[X_1].$$

Therefore when n tends to ∞ , the random variable $\hat{\mu}$ concentrates around μ with a standard deviation of order $1/\sqrt{n}$. This speed is arguably too slow for a lot of applications, that is why we will present in this subsection other methods to devise efficient estimators.

When n is big enough, the central limit Theorem provides a limiting confidence interval.

Theorem 5.14 (Central limit). *Given a sequence of independent identically distributed random variables $(X_i)_{i \in \mathbb{N}} : \Omega \rightarrow \mathbb{R}^N$ such that $(\forall i \in \mathbb{N}) \mathbb{E}[X_i] = \mu$ and $\mathbb{E}[(X_i - \mu)^2] = \sigma^2 \leq \infty$ then for all $t \in \mathbb{R}$:*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\sqrt{n}}{\sigma} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) \leq t \right) = \text{erf}(t).$$

Denoting the sample (here biased) standard deviation:

$$\hat{\sigma} \equiv \sqrt{\frac{1}{n} \sum_{i=1}^n (h(X_i) - \hat{\mu})^2},$$

one has, for n big enough, the pseudo confidence interval:

$$\mathbb{P} \left(\hat{\mu} - 1.96 \frac{\hat{\sigma}}{n} \leq \mu \leq \hat{\mu} + 1.96 \frac{\hat{\sigma}}{n} \right) \geq 0.95.$$

Once again, the question is to know what does it means exactly for n to be “big enough”.

Let us give some examples to understand the issues it better.

Example 5.15. Let us consider a Cauchy random vector $X \sim \frac{1}{\pi} \frac{1}{1+x^2}$ and we want to estimate the probability:

$$\theta \equiv \mathbb{P}(X \geq 2)$$

we will provide below several estimators and compute for all of them the variance to be able to compare them. They all depend on a sampling of n i.i.d. random variables $X_1, \dots, X_n \sim \frac{1}{\pi} \frac{1}{1+x^2}$.

- $\hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i \geq 2}$. Of course $\mathbb{E}[\hat{\theta}_1] = \mathbb{P}(X \geq 2)$ and the variance computes:

$$\mathbb{V}[\hat{\theta}_1] = \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}[\mathbb{1}_{X_i \geq 2}] = \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}[\mathbb{1}_{X_i \geq 2}^2] - \mathbb{E}[\mathbb{1}_{X_i \geq 2}]^2 = \frac{1}{n}(\theta - \theta^2),$$

since $\mathbb{1}_{X_i \geq 2}^2 = \mathbb{1}_{X_i \geq 2}$. Numerically (integrating the Cauchy distribution), that gives us $\mathbb{V}[\hat{\theta}_1] \approx \frac{0.120}{n}$.

- Noting that $\theta = \frac{1}{\pi} \int_2^{+\infty} \frac{1}{1+x^2} = \frac{1}{2\pi} \left(\int_{-\infty}^{-2} \frac{1}{1+x^2} + \int_2^{+\infty} \frac{1}{1+x^2} \right)$, one may rather choose $\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{|X_i| \geq 2}$. Once again, $\mathbb{E}[\hat{\theta}_2] = \mathbb{P}(X \geq 2) = \theta$ and the variance here computes:

$$\mathbb{V}[\hat{\theta}_2] = \frac{1}{4n} \mathbb{E}[\mathbb{1}_{|X_i| \geq 2}] - \mathbb{E}[\mathbb{1}_{|X_i| \geq 2}]^2 = \frac{1}{2n}(\theta - 4\theta^2),$$

That gives us $\mathbb{V}[\hat{\theta}_2] \approx \frac{0.58}{n} \leq \mathbb{V}[\hat{\theta}_1]$.

- Going further, we note that $\theta = \frac{1}{2} \left(1 - \frac{1}{\pi} \int_{-2}^2 \frac{1}{1+x^2} \right)$ and then rather choose $\hat{\theta}_3 \equiv \frac{1}{2} \left(1 - \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{|X_i| \leq 2} \right)$. With this choice one can reduce a bit more the variance and obtain $\mathbb{V}[\hat{\theta}_3] \approx \frac{0.002}{n}$.
- Finally with the change of variable $x \rightarrow \frac{1}{x}$, one has the new identity $\theta = \frac{1}{\pi} \int_0^{\frac{1}{2}} \frac{1/x^2}{1+1/x^2} dx = \frac{1}{\pi} \int_0^{\frac{1}{2}} \frac{1}{1+x^2} dx$ and the associated estimator would be $\hat{\theta}_4 \equiv \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{0 \leq X_i \leq \frac{1}{2}}$. Here an estimation of the variance gives us $\mathbb{V}[\hat{\theta}_4] \approx \frac{10^{-4}}{n}$.

With those example we saw that on practical examples, it is generally possible to find some small tricks to find estimator with comparatively small standard deviation (from $\hat{\theta}_1$ to $\hat{\theta}_4$ the standard deviation reduces by a factor $\sqrt{10^3} \approx 32$ – that means that one needs 32 times less samples to estimate θ with the same precision error). We will see in next subsection some more systematic techniques to optimize estimators

3.4 Importance sampling

Considering a measurable mapping $h : \mathbb{R} \rightarrow \mathbb{R}$ and a density $f : \mathbb{R} \rightarrow [0, 1]$, we try to estimate $\int_{\mathbb{R}} h(x)f(x)dx$. Importance sampling relies on the following identity, true for any $g : \mathbb{R} \rightarrow \mathbb{R}$, measurable and such that $g(x) = 0 \implies f(x) = 0$ (the measure induced by f is absolutely continuous under the measure induced by g):

$$\int_{\mathbb{R}} h(x)f(x)dx = \int_{\mathbb{R}} \frac{h(x)f(x)}{g(x)}g(x)dx.$$

The idea is then to play on g in order to find the best estimator possible. The easiest way to compare estimators is found through a comparison of the variance, and the optimal estimator to that respect is given by next theorem.

Theorem 5.16 (Rubenstein). Given a measurable mapping $h : \mathbb{R} \rightarrow \mathbb{R}$ and a density $f : \mathbb{R} \rightarrow \mathbb{R}_+$, the density $g : \mathbb{R} \rightarrow \mathbb{R}_+$ that minimizes the variance of $\frac{h(X)f(X)}{g(X)}$ for $X \sim g$ is:

$$g^* : x \mapsto \frac{|h(x)|f(x)}{\int_{\mathbb{R}} |h(t)|f(t)dt}.$$

This theorem is proven thanks to Jensen inequality.

Lemma 5.17 (Jensen inequality). *Given a convex mapping $\phi : \mathbb{R} \rightarrow \mathbb{R}$ and a random variable $X : \Omega \rightarrow \mathbb{R}$ one has the inequality:*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)].$$

For concave mappings $\phi : \mathbb{R} \rightarrow \mathbb{R}$, one has the converse identity.

Proof of Theorem 5.16 Given a random variable $X \sim g$, one can first note that:

$$\mathbb{V}\left[\frac{h(X)f(X)}{g(X)}\right] = \mathbb{E}\left[\left(\frac{h(X)f(X)}{g(X)}\right)^2\right] - \mathbb{E}\left[\frac{h(X)f(X)}{g(X)}\right]^2.$$

The term $\mathbb{E}\left[\frac{h(X)f(X)}{g(X)}\right] = \int_{\mathbb{R}} h(x)f(x)dx$ is independent of g , one therefore just tries to minimize the first term that satisfies thanks to Jensen inequality:

$$\mathbb{E}\left[\left(\frac{h(X)f(X)}{g(X)}\right)^2\right] \geq \mathbb{E}\left[\frac{|h(X)f(X)|}{g(X)}\right]^2.$$

To conclude, first remark that $\mathbb{E}\left[\frac{|h(X)f(X)|}{g(X)}\right]^2$ is independent of g and that it is equal to $\mathbb{E}\left[\left(\frac{h(X)f(X)}{g(X)}\right)^2\right]$ when $g = g^*$. That exactly implies that g^* is the density that minimize the variance. Be careful that $x \mapsto \frac{h(x)f(x)}{\int_{\mathbb{R}} h(t)f(t)dt}$ is not a density, we need to take the absolute value of $h(x)$ to ensure that $g^*(x)$ is positive. $\square$

4 Conditioning and Bayesian statistics

4.1 Independence

We are still placed in a probability space $(\omega, \mathcal{F}, \mathbb{P})$.

Definition 6. *Two events $A, B \in \mathcal{F}$ are said to be independent iif:*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Remark 5.18. • *Events $A \in \mathcal{F}$ of null probability are independent with any other events $B \in \mathcal{F}$ since $\mathbb{P}(A \cap B) \leq \mathbb{P}(A) = 0$*

- *Two distinct events of non null probability are not independent since $\mathbb{P}(A)\mathbb{P}(B) \neq 0 = \mathbb{P}(A \cap B)$*
- *Let us consider the case $\Omega = [0, 1] \times [0, 1]$ and $\mathbb{P}$ is uniformly distributed on Ω . Given $A \in \mathcal{F}$, $\mathbb{P}(A)$ is exactly the area of A . Given any intervals $[x, x+a], [y, y+b] \subset [0, 1]$, the events $A = [x, x+a] \times [0, 1]$ and $B = [0, 1] \times [y, y+b]$ are independent since one has:*

$$\mathbb{P}(A \cap B) = a \cdot b = (a \cdot 1)(1 \cdot b) = \mathbb{P}(A)\mathbb{P}(B).$$

The independence between random variables relies on the notion of independence of events.

Definition 7. *Two random variables $X : \Omega \rightarrow \mathbb{R}^p$ and $Y : \Omega \rightarrow \mathbb{R}^q$ are said to be independent iif for all Borel sets $A \in \mathcal{B}(\mathbb{R}^p)$ and $B \in \mathcal{B}(\mathbb{R}^q)$, $X^{-1}(A)$ and $Y^{-1}(B)$ are independent.*

Given an event $A \in \mathcal{F}$, we say that X is independent with A if A is independent with $X^{-1}(E)$ for all $E \in \mathcal{B}(\mathbb{R})$.

Example 5.19. • *We already presented in Example 5.2 two non independent random variables when $\Omega = [0, 2\pi]$, $X : \omega \mapsto \cos(\omega)$ and $Y : \omega \mapsto \sin(\omega)$.*

- *To present a simple example of two independent random variables, one can consider again the uniform probability on $\omega = [0, 1] \times [0, 1]$ setting $X : (\omega_1, \omega_2) \mapsto \omega_1$ and $Y : (\omega_1, \omega_2) \mapsto \omega_2$. Then for any intervals $[x, x+a], [y, y+b] \subset [0, 1]$, one knows that $X^{-1}([x, x+a]) = [x, x+a] \times [0, 1]$ and $Y^{-1}([y, y+b]) = [0, 1] \times [y, y+b]$ are independent. Since by definition of the Borel space, all events are union, intersections or complementary of intervals, one can deduce that the independence between $X^{-1}(A)$ and $Y^{-1}(B)$ generalizes to any Borel set $A, B \in \mathcal{B}([0, 1])$.*